

# Chapter 32

## An Introduction to Animal Diversity

PowerPoint® Lecture Presentations for

### **Biology**

*Eighth Edition*

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# Animals Overview

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- Animals are **multicellular, heterotrophic eukaryotes** with **tissues** that develop **from embryonic layers**.
- There are exceptions to nearly every criterion for distinguishing animals from other life-forms.
- 1.3 million living species of animals have been identified.

Which of these organisms are animals?



# Animal Structure and Specialization

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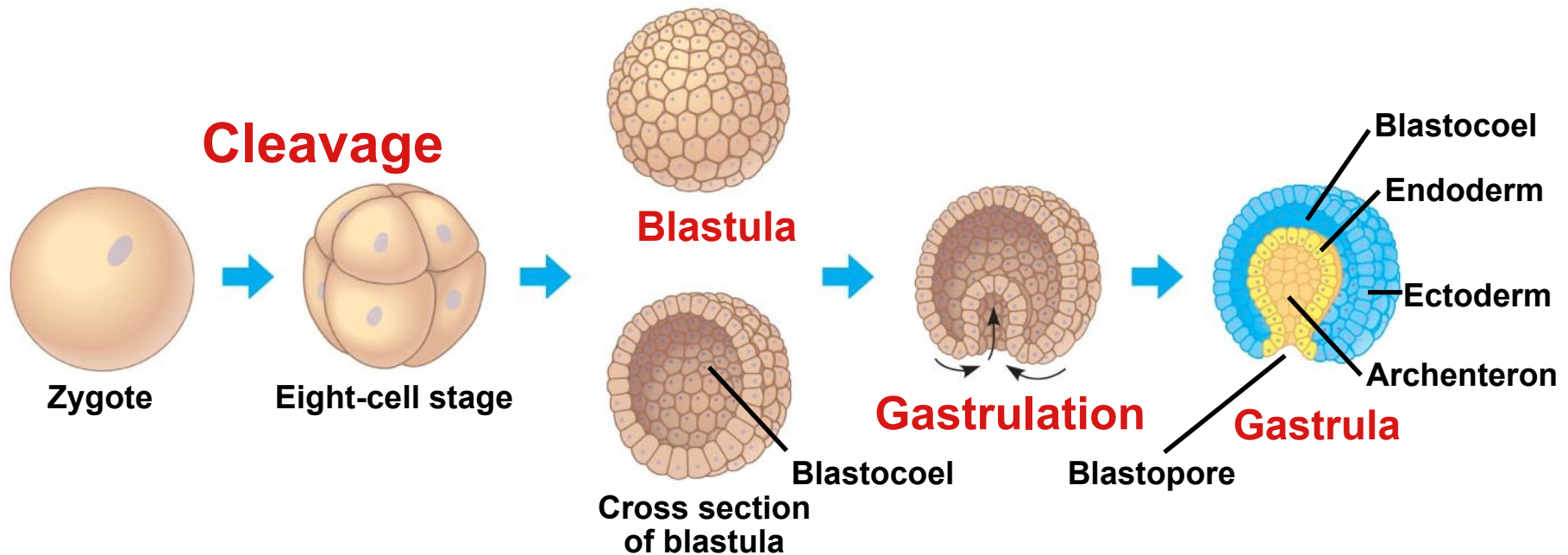
- *Nutritional Mode*: Animals are *heterotrophs* that *ingest* their food.
- Animals are *multicellular* eukaryotes.
- Their cells *lack cell walls*.
- Their bodies are held together by *structural proteins* such as *collagen*.
- *Nervous tissue* and *muscle tissue* are unique to animals.

# Reproduction and Development

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- Most animals *reproduce sexually*, with the *diploid stage* usually *dominating* the life cycle.
- *After fertilization*, the *zygote* undergoes rapid cell division called *cleavage*.
- Cleavage leads *to* formation of a *blastula*.
- The blastula undergoes *gastrulation*, forming a *gastrula* with different *layers* of *embryonic tissues*.

# Animal Early Embryonic Development



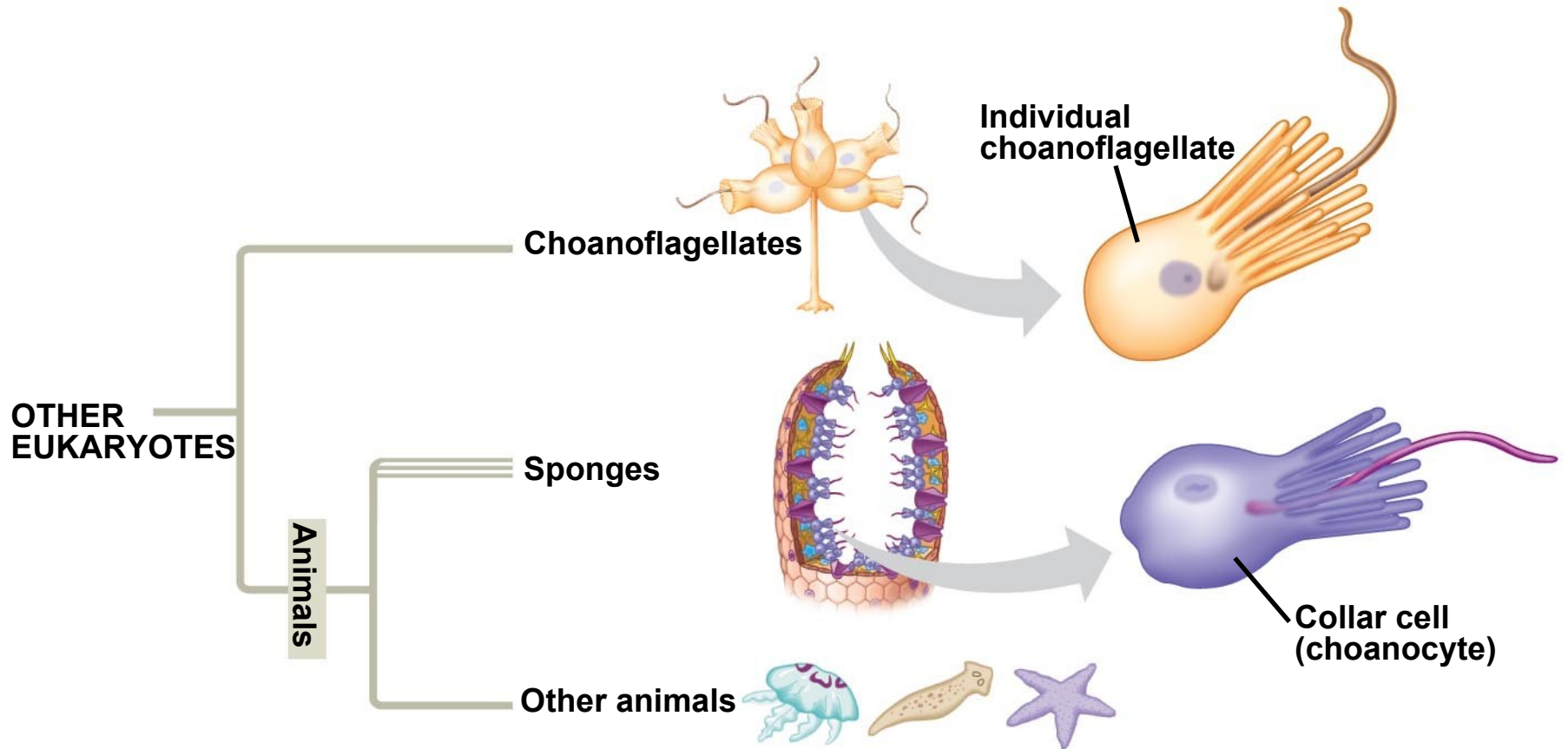
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- Many animals have at least one larval stage.
  - A *larva* is sexually immature and morphologically distinct from the adult; it eventually undergoes *metamorphosis*.
  - *All animals*, and only animals, have **Hox genes** that *regulate* the development of *body form*.
  - Although the *Hox* family of genes has been highly conserved, it can produce a wide diversity of *animal morphology*.

# The history of animals spans more than half a billion years

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- The animal kingdom includes a great diversity of living species and an even greater diversity of extinct ones.
- The *common ancestor* of living animals may have lived between 675 and 875 million years ago.
- This ancestor may have resembled modern *choanoflagellates*, *protists* that are the closest living relatives of animals.

# Three lines of evidence that choanoflagellates **protists** are **closely related to animals**



Early members of the animal fossil record include the Ediacaran biota, which dates from 565 to 550 million years ago

| 1.5 cm |



(a) *Mawsonites spriggi*

0.4 cm  
|



(b) *Spriggina floundersi*

# Paleozoic Era (542–251 Million Years Ago)

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- The ***Cambrian explosion*** (535 to 525 million years ago) marks the earliest fossil appearance of *many major groups of living animals*.
- There are several hypotheses regarding the cause of the Cambrian explosion
  - *New predator-prey relationships*
  - *A rise in atmospheric oxygen*
  - The evolution of the ***Hox gene*** complex.

# A Cambrian seascape



# Mesozoic Era (251–65.5 Million Years Ago)

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- Animal diversity continued to increase through the Paleozoic, but was punctuated by mass extinctions.
- *Animals* began to make an *impact on land* by 460 million years ago.
- Vertebrates made the transition to land around 360 million years ago.
- *Coral reefs* emerged, becoming important marine ecological niches for other organisms.
- *During the Mesozoic era, dinosaurs were the dominant terrestrial vertebrates.*
- The *first mammals* emerged.

# Cenozoic Era (65.5 Million Years Ago to the Present)

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- The beginning of the Cenozoic era followed mass extinctions of both terrestrial and marine animals.
- These extinctions included the large, nonflying dinosaurs and the marine reptiles.
- *Modern mammal orders and insects diversified during the Cenozoic.*

# Animals can be characterized by “body plans”

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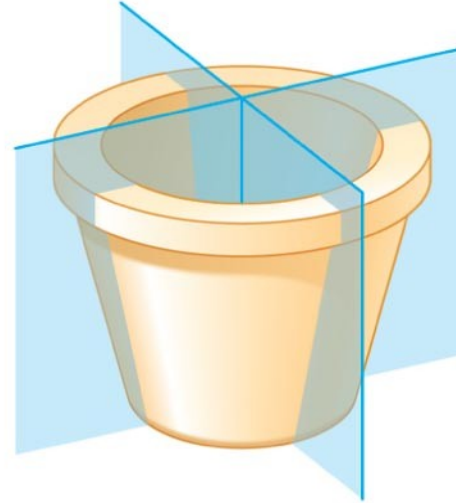
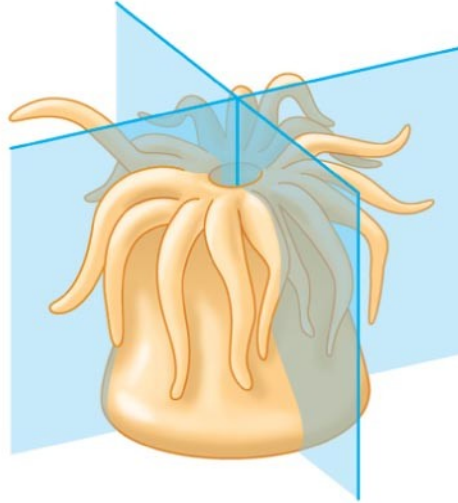
- Zoologists sometimes categorize animals according to a **body plan**, a set of morphological and developmental traits.
- A **grade** is a group whose members *share key biological features*.
- A grade is not necessarily a *clade*, or monophyletic group.

# Body Plan -- Symmetry

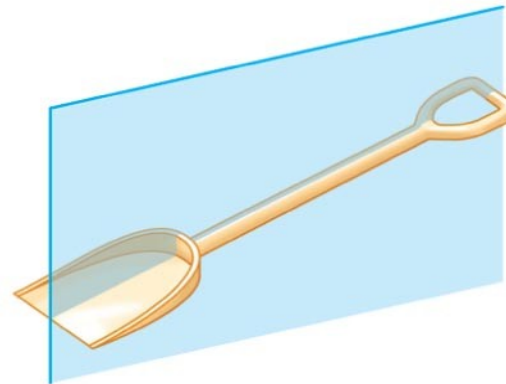
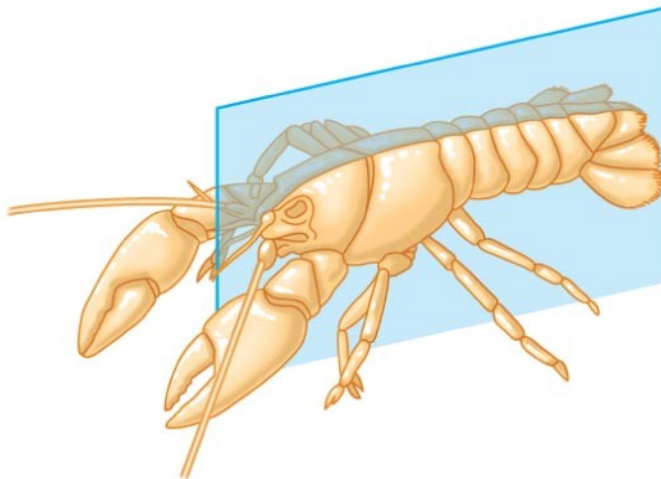
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- Animals can be categorized according to the symmetry of their bodies, or lack of it.
- Some animals have *radial symmetry*.
- Two-sided symmetry is called bilateral symmetry.
- Animals with *bilateral symmetry* have:
  - A **dorsal** (top) side and a **ventral** (bottom) side
  - A right and left side
  - **Anterior** (head) and **posterior** (tail) ends
  - *Cephalization*, the development of a head. (*Brain...*)

# Animal Body Symmetry



(a) **Radial** symmetry



(b) **Bilateral** symmetry

# Body Plan -- Tissues

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- Animal body plans also vary according to the organization of the animal's tissues.
- Tissues are collections of specialized cells isolated from other tissues by membranous layers.
- *During development, **three germ layers** give rise to the tissues and organs of the animal embryo.*

# *Embryonic Germ Layers*

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- ***Ectoderm*** is the germ layer *covering* the embryo's surface.
- ***Endoderm*** is the *innermost* germ layer and lines the *developing digestive tube*, called the *archenteron*.
- **Diploblastic** animals have ectoderm and endoderm.
- **Triploblastic** animals also have a *middle mesoderm* layer; these include all bilaterians.

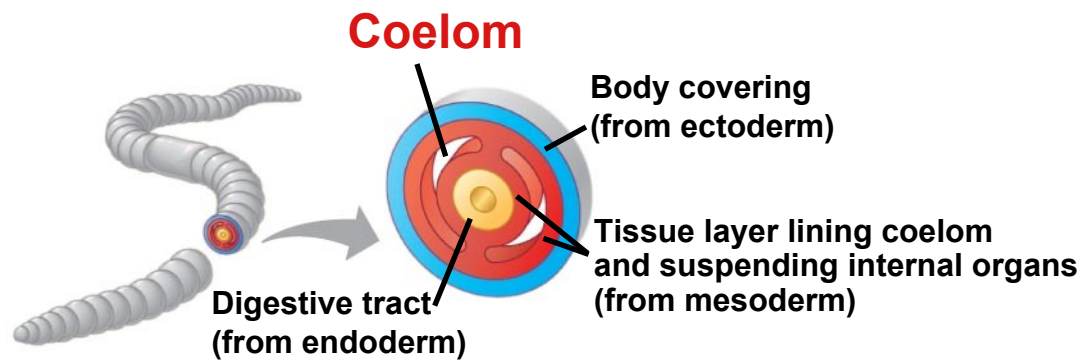
# **Body Cavities**

*Most triploblastic animals possess a body cavity.*

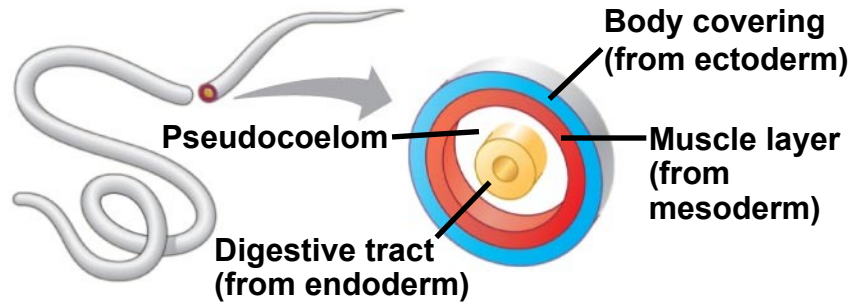
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- A *true body cavity* is called a **coelom** and is derived from *mesoderm*. **Coelomates** are animals that possess a true coelom.
- A pseudocoelom is *a body cavity* derived from the *mesoderm* and *endoderm*. Triploblastic animals that possess a pseudocoelom are called **pseudocoelomates**.
- Triploblastic animals that *lack a body cavity* are called **acoelomates**.

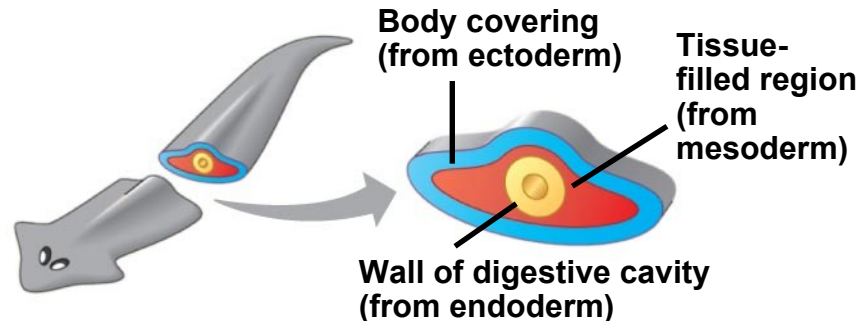
# Triploblastic Animals **Body Cavities**



(a) **Coelomate - true body cavity**



(b) **Pseudocoelomate**



(c) **Acoelomate - lack a body cavity**

# *Cleavage:*

## *protostome or deuterostome development*

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- In protostome development, cleavage is **spiral** and **determinate**.
- In deuterostome development, cleavage is **radial** and **indeterminate**.
- With *indeterminate cleavage*, **each cell** in the early stages of cleavage *retains the capacity to develop into a complete embryo*.
- Indeterminate cleavage makes possible identical twins, and embryonic stem cells.

## Protostome

Development  
molluscs, annelids

Eight-cell stage

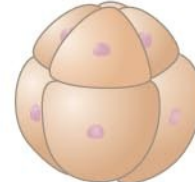


**Spiral and determinate**

## Deuterostome

Development  
echinoderm, chordates

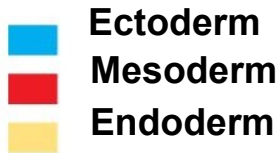
Eight-cell stage



**Radial and indeterminate**

(a) **Cleavage**

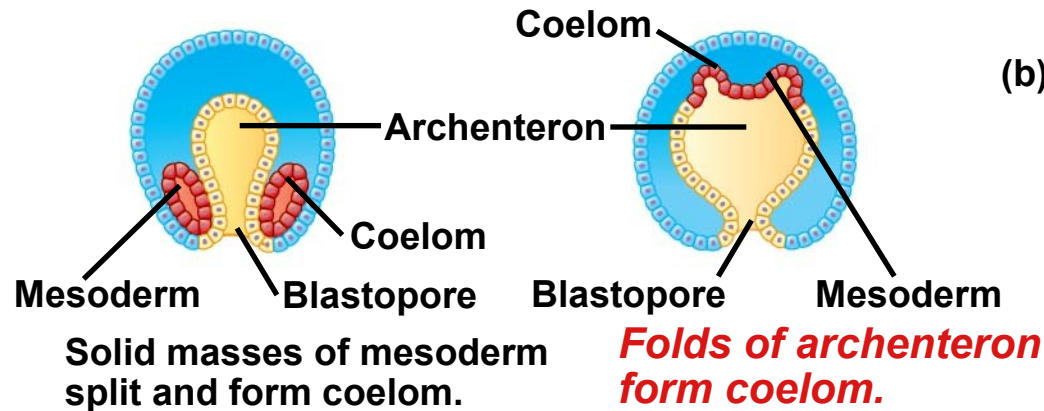
### Key



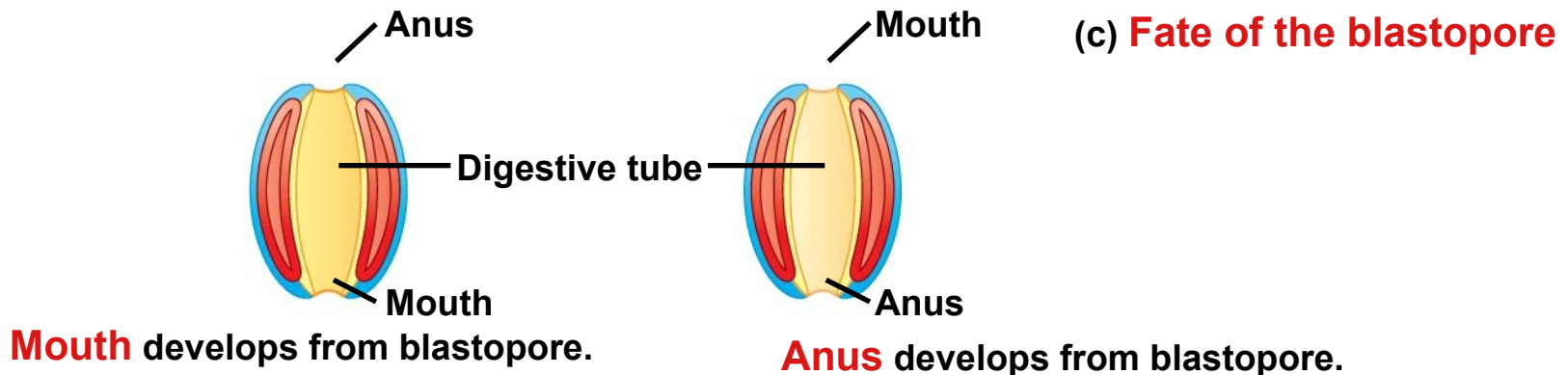
Ectoderm

Mesoderm

Endoderm



(b) **Coelom formation**

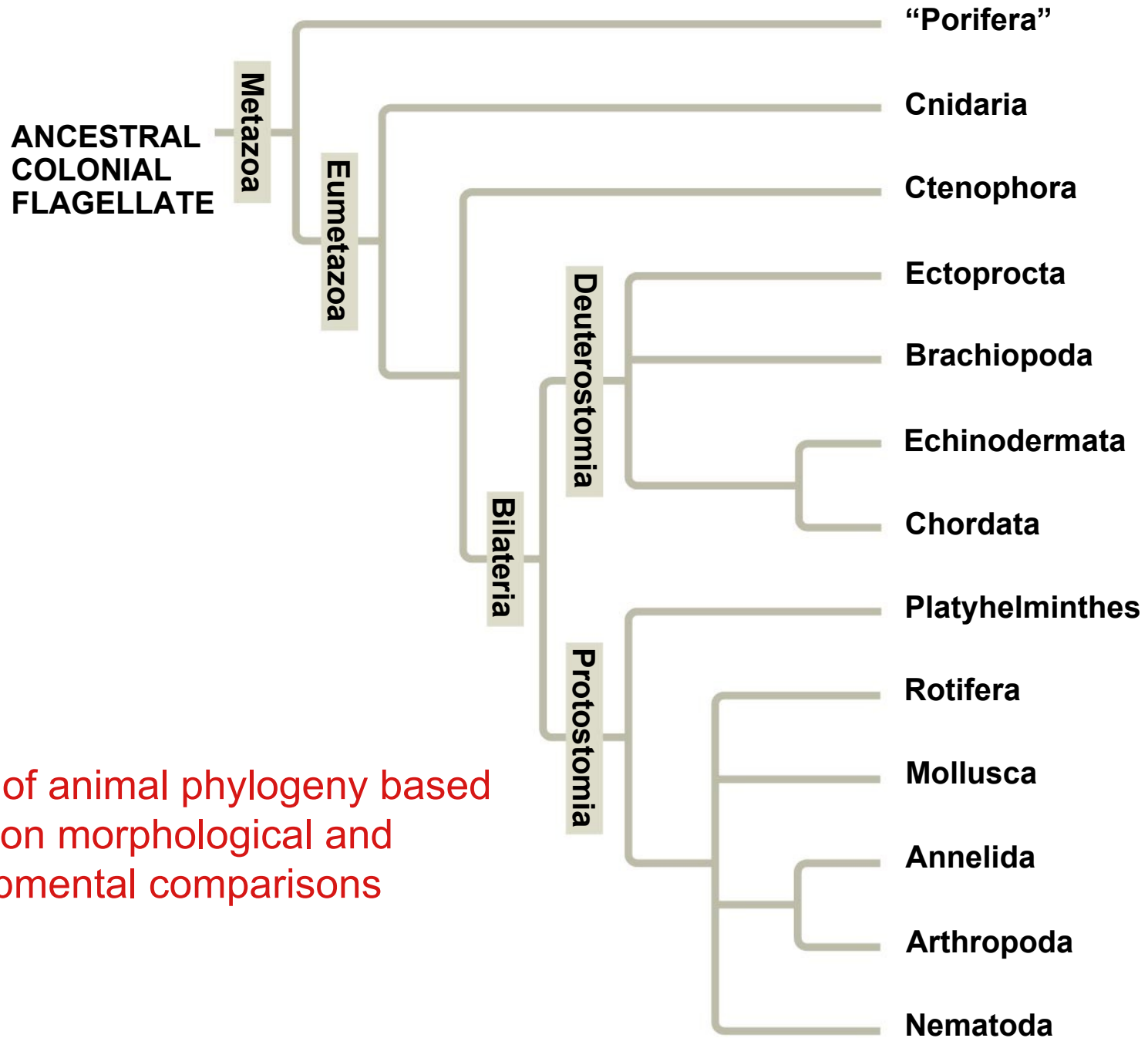


(c) **Fate of the blastopore**

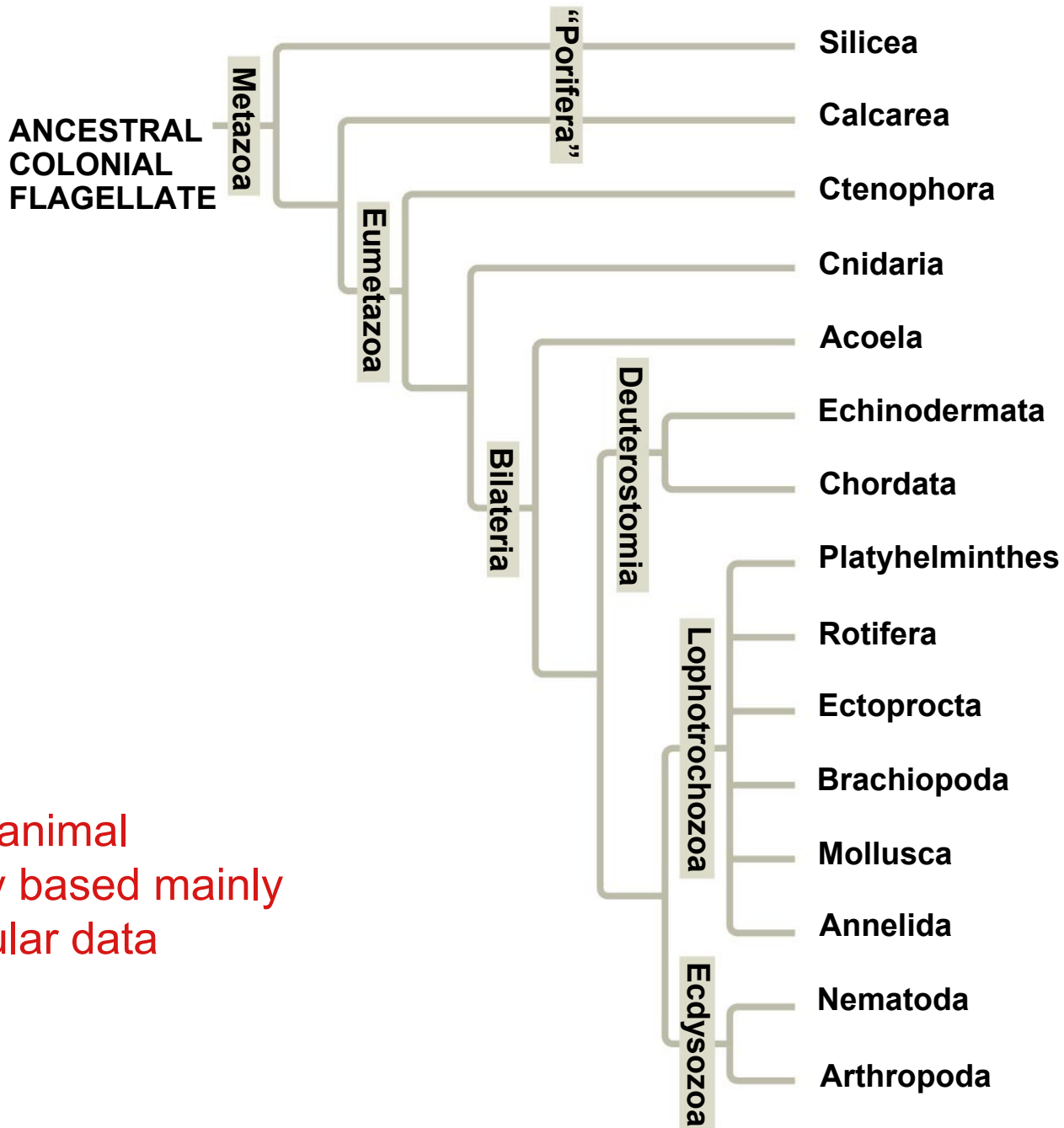
# New views of animal phylogeny are emerging from molecular data

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- Zoologists recognize about three dozen animal phyla.
- Current debate in animal systematics has led to the development of two phylogenetic hypotheses, but others exist as well.
- One hypothesis of animal phylogeny is based mainly on morphological and developmental comparisons.
- Another hypothesis is based mainly on molecular data.



A view of animal phylogeny based mainly on morphological and developmental comparisons



A view of animal  
phylogeny based mainly  
on molecular data

# Points of Agreement

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- All animals share a *common ancestor*.
- Sponges are basal animals.
- Eumetazoa is a clade of animals - *eumetazoans* with *true tissues*.
- Most animal phyla belong to the clade Bilateria, and are called *bilaterians*.
- Chordates and some other phyla belong to the clade Deuterostomia.

# Progress in Resolving Bilaterian Relationships

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- The morphology-based tree divides bilaterians into two clades: deuterostomes and protostomes.
- In contrast, recent molecular studies indicate three bilaterian clades: Deuterostomia, Ecdysozoa, and Lophotrochozoa.
- ***Ecdysozoans shed their exoskeletons through a process called ecdysis.***

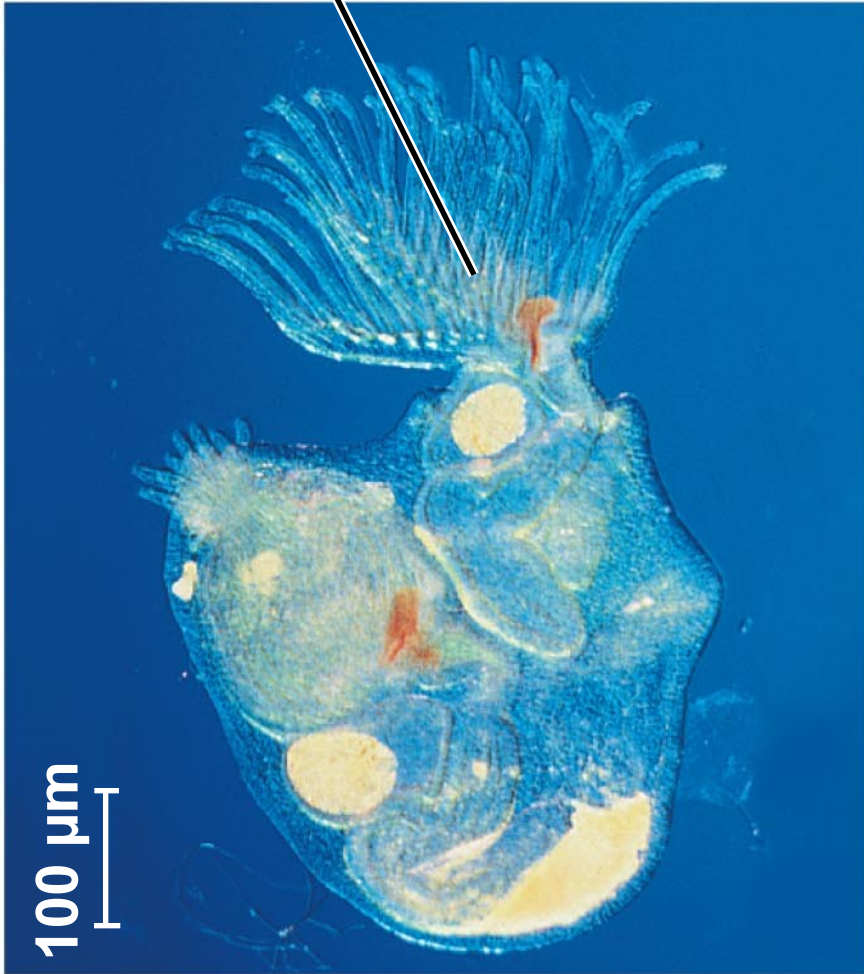
# Ecdysis - Shedding of Exoskeleton



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- Some **lophotrochozoans** have a feeding structure called a **lophophore**.
  - Other phyla go through a distinct developmental stage called the **trochophore larva**.

# Lophotrochozoans Characteristics

Lophophore

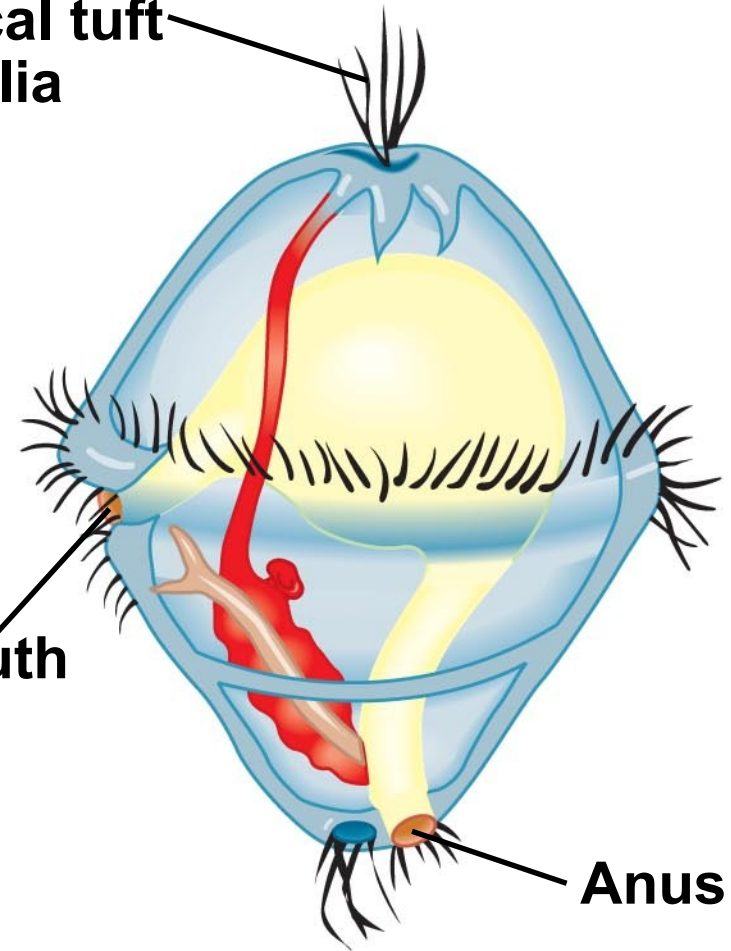


(a) An ectoproct

Apical tuft  
of cilia

Mouth

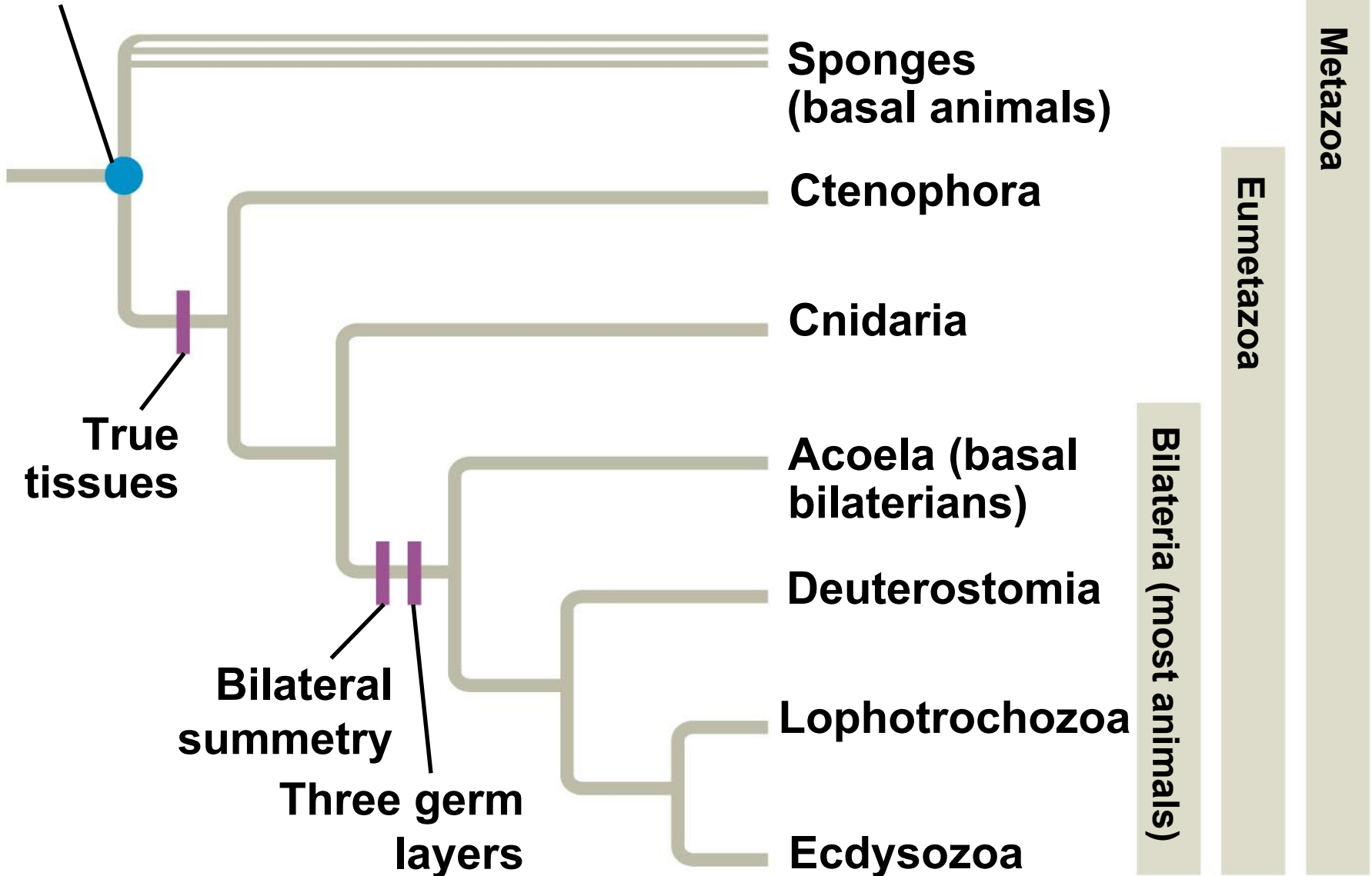
Anus



(b) Structure of a trochophore larva

# Animal Phylogeny

Common ancestor  
of all animals



# You should now be able to:

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1. List the characteristics that combine to define animals.
  2. Summarize key events of the Paleozoic, Mesozoic, and Cenozoic eras.
  3. Distinguish between the following pairs or sets of terms:  
radial and bilateral symmetry;  
diploblastic and triploblastic;  
spiral and radial cleavage;  
determinate and indeterminate cleavage;  
acoelomate, pseudocoelomate, and coelomate
  4. Compare the developmental differences between protostomes and deuterostomes.
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